分数 3 作者 朱建科 单位 浙江大学

Given a tree of degree 3, if there are 49 nodes of degree 2, and 25 nodes of degree 3, then how many leaf nodes are there?

A. 101 B. 100 C. 150 D. 50

评测结果 答案正确

得分 3 分

R2-2 分数 3 作者 朱建科 单位 浙江大学

Which of the following statements is correct regarding a binary search tree storing unique integers?

A. The integers 8 and 9 may be at the same level.

B. If 8 and 10 are at the same level, then 9 must be their parent node.

C. If 8 and 10 are at the same level, then 9 must not be their parent node.

D. None.

评测结果 答案正确

得分 3 分

R2-3 分数 3 作者 朱建科 单位 浙江大学

For a binary tree, if its post-order travel sequence is { 1, 3, 2, 6, 5, 7, 4 }, and its in-order travel sequence is { 1, 2, 3, 4, 5, 6, 7 }, then which of the following statement is TRUE?

A. This is a complete binary tree. B. 6 is the parent of 5.

C. This is a binary search tree. D. 3 is the parent of 2.

评测结果 答案正确

得分 3 分

R2-4 分数 3 作者 陈超超 单位 浙江大学

In a doubly linked circular list, how many pointers must be updated to insert q after p?

A. 1 B. 2 C. 3 D. 4

评测结果 答案正确

得分 3 分

R2-5 分数 3 作者 陈超超 单位 浙江大学

Design an algorithm to determine whether the parentheses `()` in an expression are properly matched. Which data structure is the most suitable for this task?

A. Sequentially stored linear list B. Queue C. Stack D. Heap

评测结果 答案正确

得分 3 分

R2-6 分数 3 作者 陈超超 单位 浙江大学

Suppose that the level-order traversal sequence of a min-heap is { 5, 18, 29, 23, 64, 47, 36}. After calling DeleteMin twice, what will be the post-order traversal of the resulting heap after it has been adjusted?

A. 47, 36, 64, 23, 29 B. 47, 64, 36, 23, 29 C. 47, 64, 29, 36, 23 D. 47, 64, 36, 29, 23

评测结果 答案正确

得分 3 分

R2-7 分数 3 作者 陈超超 单位 浙江大学

It is known that a 3-heap is a heap whose nodes have 3 children. In a min-3-heap with 2025 elements, which one of the following indices is NOT a possible position for the maximum key?

A. 675 B. 676 C. 1012 D. 1013

评测结果 答案正确

得分 3 分

R2-8 分数 3 作者 朱建科 单位 浙江大学

For an in-order threaded binary tree, if the post-order and in-order traversal sequences are FDEBGCA and FDBEACG, respectively. Which pair of nodes' left links are both threads?

(●) A. E and G. ( ) B. B and G. ( ) C. C and D. ( ) D. D and E.

评测结果 答案正确

得分 3 分

R2-9 分数 3 作者 朱建科 单位 浙江大学

Consider the following operations performed on an initially empty Binary Search Tree (BST):
Insert: 50, 30, 70, 20, 40, 60, 80
Delete: 50 (replaced with its in-order successor)
Insert: 55
After these operations, which node will be the in-order predecessor of 60 in the final BST?

( ) A. 30.

( ) B. 40.

(●) C. 55.

( ) D. The in-order predecessor does not exist for 60.

评测结果 答案正确

得分 3 分

R2-10 分数 3 作者 陈佳伟 单位 浙江大学

The array representation of the disjoint sets is given by { 3, 3, 4, 7, 4, -3, -6, 6, 8, -2, 10 }. Keep in mind that the elements are numbered from 1 to 11, and the depth of the root is 0. After invoking `Union(Find(9), Find(11))`, `Union(Find(11), Find(3))` with union-by-size and path compression, what is the maximum depth of all the nodes?

( ) A. 1 (●) B. 2 ( ) C. 3 ( ) D. 4

评测结果 答案正确

得分 3 分

R2-11 分数 3 作者 陈佳伟 单位 浙江大学

Which of the following statements about hashing is incorrect?

- A. In hashing with quadratic probing to resolve collisions, a new element can definitely be inserted if the table size is 14 and 7 cells are occupied.
- B. If an insertion into the hash table fails, rehashing is necessary, and the new table size is about twice the original size.
- C. If in double hashing $\mathrm{Hash}_2(k) = k^2$, then double hashing effectively becomes quadratic probing.
- D. the expected number of probes for insertions is equal to unsuccessful searches in linear probing method.

评测结果 答案错误

得分 0 分

R2-12 分数 3 作者 陈佳伟 单位 浙江大学

Insert {26,30,13,17,40,21,22,10} one by one into an initially empty hash table of size 11 with the hash function $H(Key) = Key\%11$, and linear probing is used to resolve collisions. After Insertion, what is the expected number of probes for an unsuccessful search?

A. $\frac{28}{11}$ B. $\frac{25}{11}$ C. $\frac{27}{11}$ D. $\frac{24}{11}$

评测结果 答案错误

得分 0 分

R2-13 分数 3 作者 陈翔 单位 浙江大学

To compute the shortest path in a weighted graph, Dijkstra's algorithm using a min-priority queue is a popular choice. For this algorithm, what is the total time spent on running the relaxation step "decrease the distance of each vertex adjacent from the current visited one"?

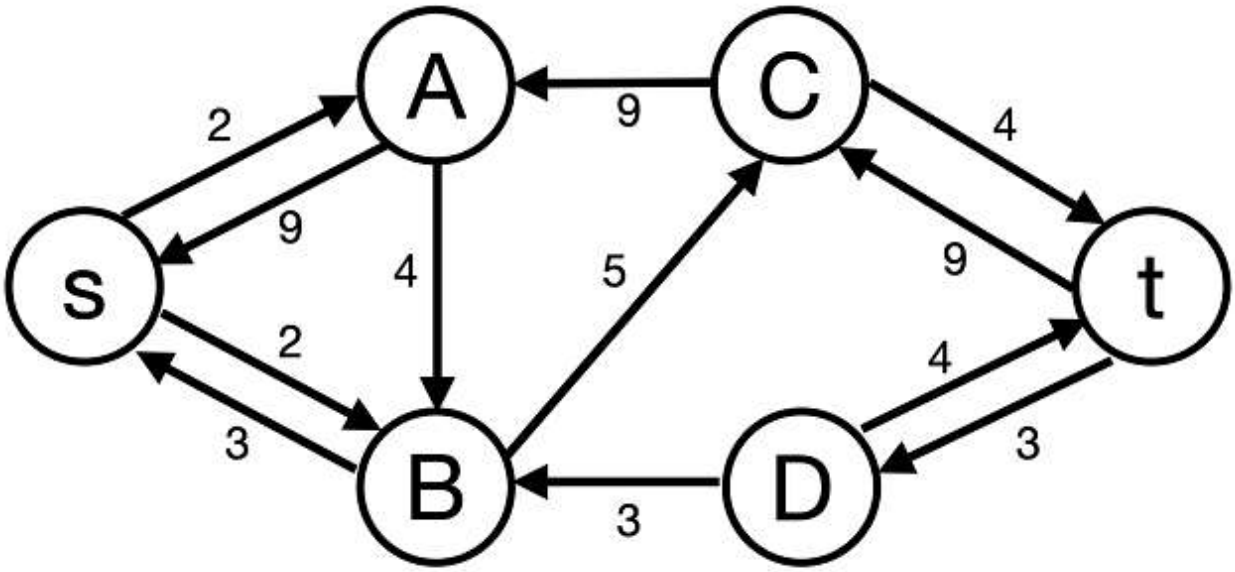
A. $O(|E|)$ B. $O(|V| * |E|)$ C. $O(|V| * \log|V|)$ D. $O(|E| * \log|V|)$

评测结果 答案正确

得分 3 分

R2-14 分数 3 作者 陈翔 单位 浙江大学

We are executing an algorithm to find the maximum flow of a given network. Currently, we have found several augmenting paths to send 12 flow from `s` to `t`, and we show the updated residual graph in the following figure.



So what is the maximum flow we can send from `s` to `t`?

- A. 12
- B. 14
- C. 16
- D. 20

评测结果 答案正确

得分 3 分

R2-15 分数 3 作者 陈翔 单位 浙江大学

Given a weakly-connected *directed* graph, what condition ensures that an Eulerian Circuit exists?

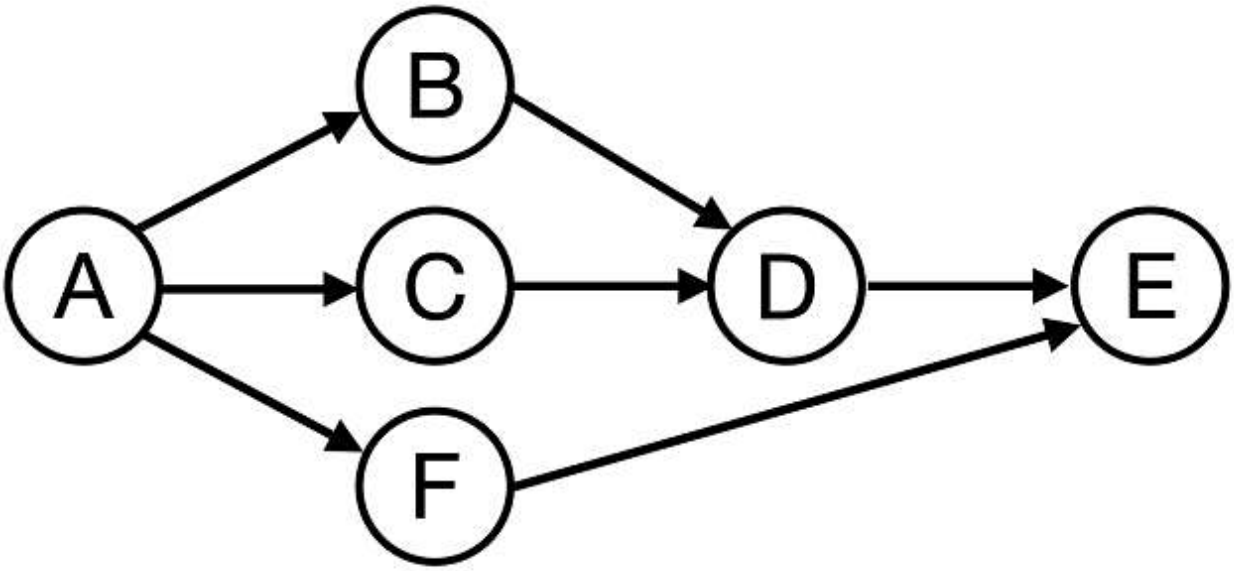
- A. All vertices have equal in-degree and out-degree.
- B. Exactly one vertex has out-degree = in-degree + 1, one vertex has in-degree = out-degree + 1, and all the rest vertices have equal in-degree and out-degree.
- C. Exactly two vertices have odd total-degree, and all the rest vertices have even total-degree.
- D. All vertices have even total-degree.

评测结果 答案正确

得分 3 分

R2-16 分数 3 作者 陈翔 单位 浙江大学

How many valid topological sorts does the following graph have?



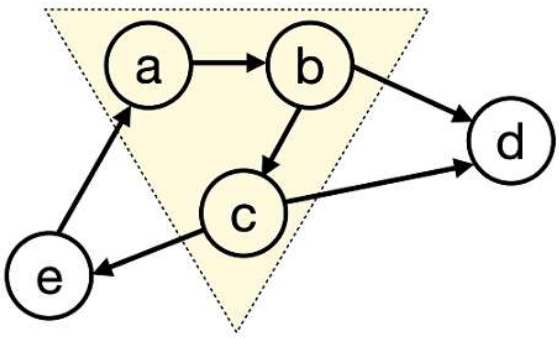
○ A. 7 ◉ B. 8 ○ C. 9 ○ D. 10

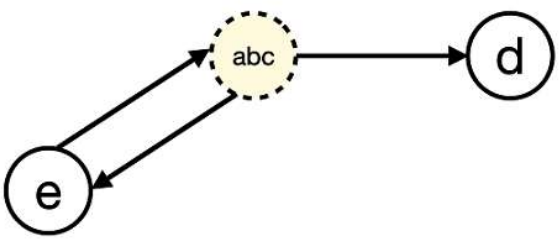
评测结果 答案正确

得分 3 分

R2-17 分数 3 作者 陈翔 单位 浙江大学

The following figure illustrates how to contract a subgraph (shown in dashed line) into a vertex.





Now given a directed graph with many *strongly connected components*, if we contract each component into a single vertex, which property does the resulting graph MUST have?

A. a complete graph　B. a disconnected graph　C. a directed acyclic graph　D. a binary tree

评测结果　答案正确

得分　3 分

R2-18　分数 3　作者 杨子祺　单位 浙江大学

Using the direct insertion sort algorithm to sort the following four lists in ascending order, the one with the fewest number of comparisons is:

A. 67,13,20,55,40,27,6,31　B. 6,13,27,20,40,31,55,67　C. 13,20,6,27,31,67,55,40　D. 55,31,40,27,6,13,67,20

评测结果　答案正确

得分　3 分

R2-19　分数 3　作者 杨子祺　单位 浙江大学

Given that the key sequence {7, 9, 14, 19, 25, 20, 16, 21} is a min-heap, after inserting the key 2 and adjusting accordingly, the resulting min-heap is:

A. {2,14,7,9,25,20,16,21,19}　B. {2,7,14,19,20,16,21,9,25}　C. {2,9,14,7,20,16,21,25,19}　D. {2,7,14,9,25,20,16,21,19}

评测结果　答案正确

得分　3 分

R2-20　分数 3　作者 杨子祺　单位 浙江大学

Which of the following options is not a possible result after the second pass of quicksort?

A. 3,7,11,9,13,17,19　B. 3,17,11,13,9,7,19　C. 7,3,11,9,17,13,19　D. 9,3,7,11,17,13,19

评测结果　答案正确

得分　3 分